Nicola Jackson

nicolacj@bu.edu | (603) 831-8964 | LinkedIn | GitHub | nicojackson.me

Education

Boston University *Expected May 2027*

Bachelor of Arts in Computer Science — GPA: 3.99 *Boston, MA*

- **Honors:** Dean's List (all semesters), CGS Merit Scholarship Recipient
- **Coursework:** Distributed Systems, Algorithms, Software Engineering, Databases, Data Structures, Computer Systems

Experience

PNC Bank *Jun. 2026 – Aug. 2026*

Technology Intern *Pittsburgh, PA*

- Engineered Aria, an internal AI coding assistant (similar to GitHub Copilot) custom-trained on proprietary systems to automate developer workflows for enterprise teams
- Authored 15+ instruction and skill files spanning API scaffolding, database connectivity (MongoDB, Kafka, Oracle), and Angular development, forming the core knowledge base driving the agent's accuracy
- Reduced repetitive developer setup tasks (boilerplate, DB configuration, environment scaffolding) by 75%, designed with a modular framework to support upcoming enterprise-wide expansion

Hack4Impact BU *May 2025 – Present*

Director of Engineering (Dec. 2025 – Present) | Lead Software Engineer (May – Dec. 2025) *Boston, MA*

- Direct 3–5 cross-functional teams (20+ engineers) delivering 3–5 full-stack applications per semester for non-profit clients
- Own end-to-end technical execution, defining system architecture, scoping project roadmaps, and conducting cross-team code reviews
- Designed the SWE hiring pipeline to evaluate 50+ applicants, having previously led a 4-person team rebuilding the chapter site using React, Next.js, and Tailwind CSS

Boston University *Aug. 2025 – Present*

Teaching Assistant (Aug. 2026 – Present) | Course Assistant (Aug. 2025 – Aug. 2026) *Boston, MA*

- Lead weekly lab sections for 200+ students, teaching core Python concepts, data structures, and object-oriented programming
- Grade midterm exams, finals, and programming assignments using standardized rubrics to provide targeted technical feedback
- Host weekly office hours to assist students with algorithmic thinking, code optimization, and real-time debugging

Projects

BankBot | ***Python, Streamlit, ChromaDB, Ollama*** *Jul. 2025*

- Built a RAG-powered banking chatbot using LlamaIndex and ChromaDB, retrieving relevant policy documents to ground LLM responses to customer questions on fees, rates, and account terms
- Integrated a locally-hosted Llama 3.2 model via Ollama with tool-calling to let the assistant query live account balances and transaction history alongside document retrieval
- Engineered keyword-triggered RAG routing and dynamic system-prompt injection (customer ID, selected account) to keep responses accurate and context-aware without unnecessary retrieval calls

Personal Portfolio | ***React, JavaScript, MongoDB, Google Gemini API*** *Mar. 2025*

- Won **1st place** in a 60-person web design competition for UI/UX quality and interactivity
- Integrated a Google Gemini API–powered chatbot and real-time GitHub activity visualization to enhance user engagement
- Deployed a dynamic, responsive portfolio site showcasing projects and live development activity

Scrappi | ***React Native, Expo, JavaScript, Node.js*** *Dec. 2025*

- Co-developed a cross-platform mobile app in a 4-person team to track household food inventory, track expiration timelines, and reduce waste
- Integrated device camera functionality for automated item logging and geospatial location services for proximity-based food resource recommendations

Study Buddy | ***Next.js, Supabase, Tailwind CSS, JavaScript*** *Aug. 2025*

- Built a full-stack productivity app featuring user authentication, task management, deadline tracking, and a Pomodoro timer
- Integrated Supabase for backend database management and user auth, leveraging Next.js server components for real-time state persistence and performance

Technical Skills

Languages	Python, Java, SQL, C, Go, JavaScript, TypeScript
Frameworks & Libraries	React, Next.js, Node.js, Express.js, Tailwind CSS, NumPy
Databases & Tools	MongoDB, PostgreSQL, Supabase, Git, GitHub, Vercel, VS Code
AI & Architecture	RAG Pipelines, Prompt Engineering, Ollama, GitHub Copilot, Claude Code